

International Airport Database

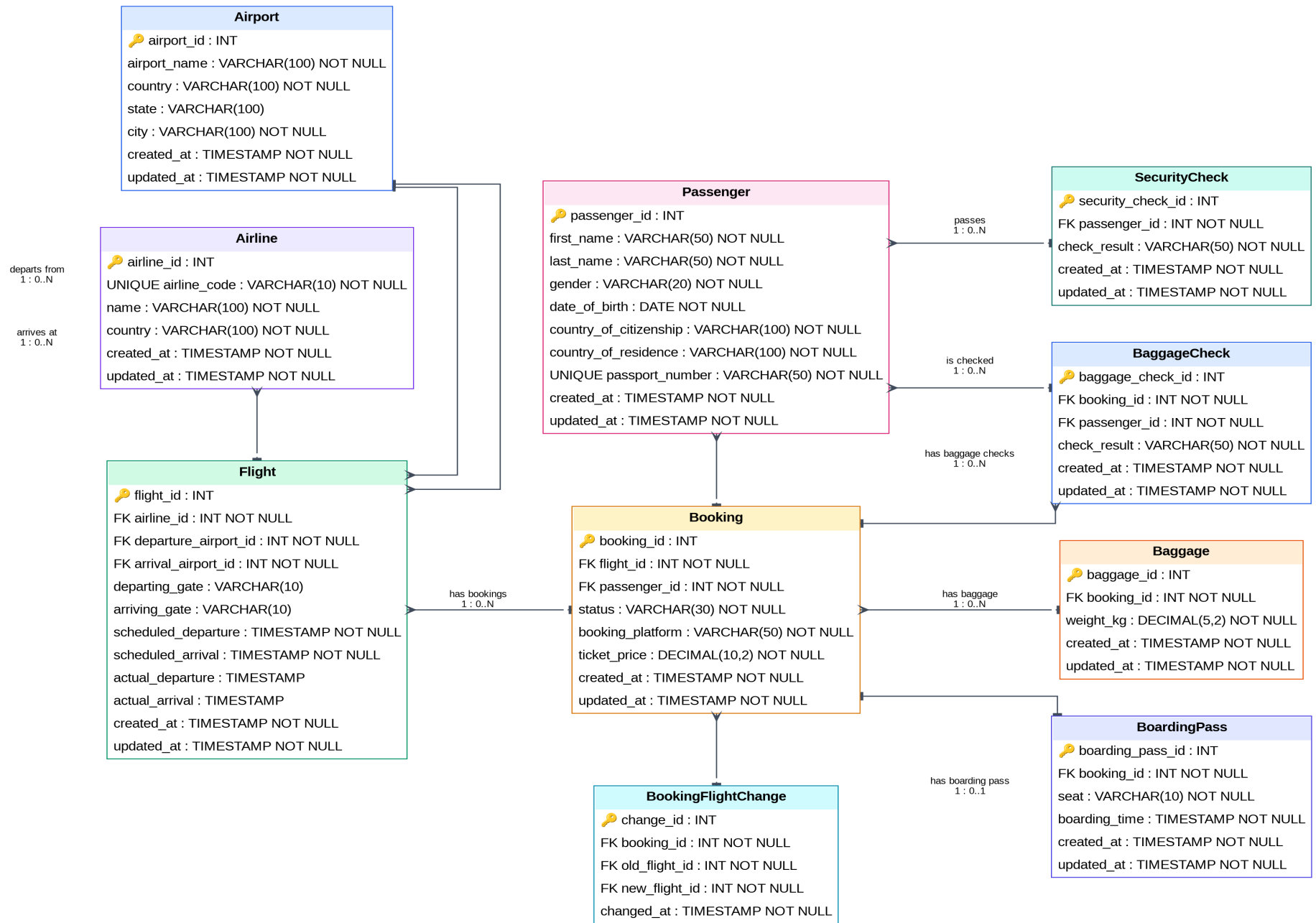
Laboratory Work 1 — ERD Diagram and Textual Description

Revised submission document

This version separates **Entities and Attributes** into two dedicated pages so every attribute, data type, and constraint is clearly readable.

1. Complete ERD Diagram

Crow's Foot-style logical ERD. PK = Primary Key, FK = Foreign Key, UNIQUE = Unique constraint, NOT NULL = required attribute.



2. Entities and Attributes — Page 1 of 2

Each entity below lists the complete attribute set from the database specification, together with the proposed SQL data type and key/constraint status.

Airport

Attribute	Data Type	Key / Constraint
airport_id	INT	PK
airport_name	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
state	VARCHAR(100)	OPTIONAL
city	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Airline

Attribute	Data Type	Key / Constraint
airline_id	INT	PK
airline_code	VARCHAR(10)	UNIQUE, NOT NULL
name	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Flight

Attribute	Data Type	Key / Constraint
flight_id	INT	PK
airline_id	INT	FK, NOT NULL
departure_airport_id	INT	FK, NOT NULL
arrival_airport_id	INT	FK, NOT NULL
departing_gate	VARCHAR(10)	OPTIONAL
arriving_gate	VARCHAR(10)	OPTIONAL

Attribute	Data Type	Key / Constraint
scheduled_departure	TIMESTAMP	NOT NULL
scheduled_arrival	TIMESTAMP	NOT NULL
actual_departure	TIMESTAMP	OPTIONAL
actual_arrival	TIMESTAMP	OPTIONAL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Passenger

Attribute	Data Type	Key / Constraint
passenger_id	INT	PK
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
gender	VARCHAR(20)	NOT NULL
date_of_birth	DATE	NOT NULL
country_of_citizenship	VARCHAR(100)	NOT NULL
country_of_residence	VARCHAR(100)	NOT NULL
passport_number	VARCHAR(50)	UNIQUE, NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Booking

Attribute	Data Type	Key / Constraint
booking_id	INT	PK
flight_id	INT	FK, NOT NULL
passenger_id	INT	FK, NOT NULL
status	VARCHAR(30)	NOT NULL
booking_platform	VARCHAR(50)	NOT NULL
ticket_price	DECIMAL(10,2)	NOT NULL

Attribute	Data Type	Key / Constraint
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

2. Entities and Attributes — Page 2 of 2

Each entity below lists the complete attribute set from the database specification, together with the proposed SQL data type and key/constraint status.

BoardingPass

Attribute	Data Type	Key / Constraint
boarding_pass_id	INT	PK
booking_id	INT	FK, NOT NULL
seat	VARCHAR(10)	NOT NULL
boarding_time	TIMESTAMP	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Baggage

Attribute	Data Type	Key / Constraint
baggage_id	INT	PK
booking_id	INT	FK, NOT NULL
weight_kg	DECIMAL(5,2)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

BaggageCheck

Attribute	Data Type	Key / Constraint
baggage_check_id	INT	PK
booking_id	INT	FK, NOT NULL
passenger_id	INT	FK, NOT NULL
check_result	VARCHAR(50)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

SecurityCheck

Attribute	Data Type	Key / Constraint
security_check_id	INT	PK
passenger_id	INT	FK, NOT NULL
check_result	VARCHAR(50)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

BookingFlightChange

Attribute	Data Type	Key / Constraint
change_id	INT	PK
booking_id	INT	FK, NOT NULL
old_flight_id	INT	FK, NOT NULL
new_flight_id	INT	FK, NOT NULL
changed_at	TIMESTAMP	NOT NULL

3. Relationships, Cardinality, and Participation

#	Relationship	Cardinality	Participation / Meaning
1	Airline → Flight (operates)	1 : 0..N	One airline may operate many flights; every Flight references one Airline.
2	Airport → Flight (departs from)	1 : 0..N	One airport may be the departure airport for many flights; every Flight has one departure airport.
3	Airport → Flight (arrives at)	1 : 0..N	One airport may be the arrival airport for many flights; every Flight has one arrival airport.
4	Passenger → Booking (makes)	1 : 0..N	A passenger may make many bookings; every Booking references one Passenger.
5	Flight → Booking (has bookings)	1 : 0..N	A flight may have many bookings; every Booking references one Flight.
6	Booking → BoardingPass	1 : 0..1	A booking may have zero or one boarding pass; a boarding pass belongs to one Booking.
7	Booking → Baggage	1 : 0..N	A booking may have no baggage or many baggage records; each Baggage belongs to one Booking.
8	Booking → BaggageCheck	1 : 0..N	A booking may have many baggage checks; each check references one Booking.
9	Passenger → BaggageCheck	1 : 0..N	A passenger may have many baggage checks; each check references one Passenger.
10	Passenger → SecurityCheck	1 : 0..N	A passenger may have many security checks; each check references one Passenger.
11	Booking → BookingFlightChange	1 : 0..N	A booking may have zero or many flight-change records; each change references one Booking.

Important: Flight has two separate foreign keys to Airport: `departure_airport_id` and `arrival_airport_id`. Therefore, two separate Airport–Flight relationships are shown.

4. Textual Description of the Solution

Airport

Stores international airport information. `airport_id` uniquely identifies each airport. The entity contains the airport name, country, state, city, and record creation/update timestamps.

Airline

Stores airline company information. `airline_id` is the primary key and `airline_code` is unique. The entity contains the airline name, country, and creation/update timestamps.

Flight

Stores scheduled and actual flight information. Each flight references one airline, one departure airport, and one arrival airport. It also stores gates, scheduled/actual times, and timestamps.

Passenger

Stores passenger profile information associated with bookings, including personal details, citizenship/residence, a unique passport number, and profile timestamps.

Booking

Represents a passenger's booking for a flight. It references one Flight and one Passenger and stores status, platform, ticket price, and timestamps.

BoardingPass

Stores boarding passes issued for bookings, including seat, boarding time, and timestamps. A booking may have zero or one boarding pass in this simplified model.

Baggage

Stores baggage registered for a booking. Each record contains weight in kilograms, the booking reference, and timestamps.

BaggageCheck

Stores baggage-check results at the airport. It references a booking and passenger and records the check result and timestamps.

SecurityCheck

Stores airport security-check results for passengers. It references a passenger and records the result and timestamps.

BookingFlightChange

Stores flight-change history for bookings in a separate table. It records the booking, old flight, new flight, and change timestamp.

5. Keys, Constraints, Legend, and Modeling Assumptions

Notation	Meaning
PK	Primary Key — uniquely identifies a record.
FK	Foreign Key — references a primary key in another entity.
UNIQUE	The value must not be duplicated.
NOT NULL	The attribute is required.
1 : 0..N	One parent can have zero or many child records.
1 : 0..1	One parent can have zero or one child record.

1. The specification mentions accounts and booking legs, but does not define their attributes or tables. They are not invented in this ERD.
2. The specification mentions aircraft information and frequent flyer programs, but provides no concrete attributes or relationships. They are treated as underspecified scope.
3. BookingFlightChange is included because the specification explicitly requires flight-booking changes to be stored in a separate table.
4. BaggageCheck follows the attributes explicitly listed in the specification. A baggage_id was not added because it was not provided.
5. The SQL data types are reasonable implementation choices. The conceptual relationships and attributes remain valid if the instructor requires different SQL types.